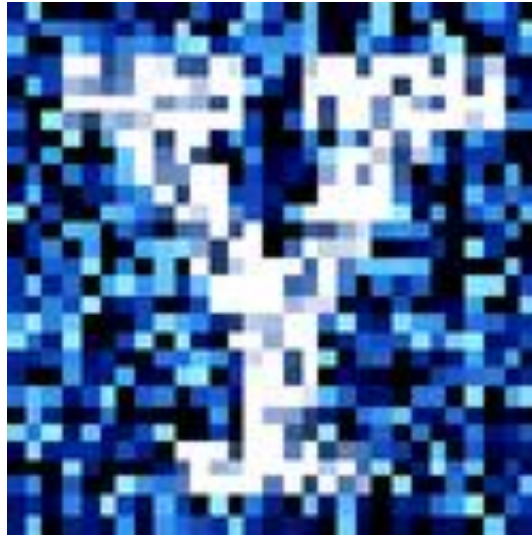


YData: Introduction to Data Science

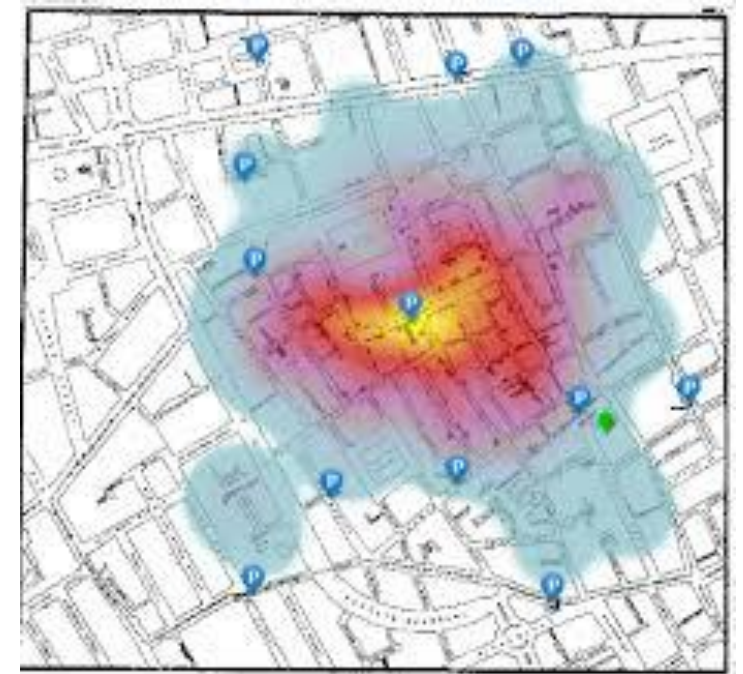


Class 15: For loops and writing functions

Overview

Quick review of pivot table and mapping

For loops and conditional statements



John Snow's ghost map

Announcement

- The midterm grading is ongoing.
 - Please email me to chat on Thu evening/Friday morning if you want to get some idea about your rough grades/standing in the class before the drop deadline.
- HW 6 is due on Gradescope on **Sunday Mar. 8th at 11pm.**
 - This is a relatively short homework.

Announcement: class project

The final project is a 6-10 pages Jupyter notebook report where you analyze your own data to address a question that you find interesting

- It's a chance to practice everything you've learned in class!

The goal of is to present a clear and compelling data analysis showing a few interesting results!

- Ideally something you are proud of that you could show off to potential future employers, etc.



Announcement: class project

Think about what questions you want to examine, find data, and load it into Python

- A few sources for data sets are listed on Canvas

You can download a project template Jupyter notebook using:

```
import YData  
YData.download_class_file('project_template.ipynb', 'home  
work')
```



Announcement: class project

A draft of the project is due on **April 10th**

- I'm telling you about this early so:
 - If you want to think/work on the project over break you can
 - If don't want to think about it over break you don't have to

There will be a one-week “peer review” period where you will give and receive feedback on three other projects

- Instructions for how to do the review will be given

The final version will be due on **April 24th**

Questions?

Collaborative projects?

Short answer: No.

- But you can discuss/brainstorm with another student on exploring a common topic of interest, possibly using different/related data or testing different hypotheses...
- Write the name of your collaborator/discussant on your project if the discussion is deemed substantial.

Homework 6 is due on Sunday March 8 11pm

Finally, I encourage everyone to continue to attend practice sessions

- Particularly if you found the midterm difficult

Where we are and where we're going...

What we have covered:

- What is Data Science
- Basics of Python (data types, lists, etc.)
- Numerical computations (numpy)
- Data tables (pandas)
- Data visualization (matplotlib and seaborn)
- Mapping

The rest of the semester:

- Functions and for loops
- Statistical analysis
- Machine Learning
- Ethics and conclusions

Quick review of pivot table and mapping

Pivot Tables and heatmaps

Pivot tables aggregate values based on two grouping variables, and create a table where:

- The rows are the levels of one cat. variable
- The columns are the levels of the second cat. variable
- The values are aggregated over a third quant. variable

```
df2 = df.pivot_table(index = "col1", columns = "col2",  
                      values = "col3", aggfunc = "max")
```

```
nba2 = nba.pivot_table(index = "TEAM", columns = "POSITION",  
                       values = "SALARY", aggfunc = "max")
```

rows columns values

	PLAYER	TEAM	POSITION	SALARY
0	De'Andre Hunter	Atlanta Hawks	SF	9.835881
1	Jalen Johnson	Atlanta Hawks	SF	2.792640
2	AJ Griffin	Atlanta Hawks	SF	3.536160
3	Trent Forrest	Atlanta Hawks	SG	0.508891
4	John Collins	Atlanta Hawks	PF	23.500000

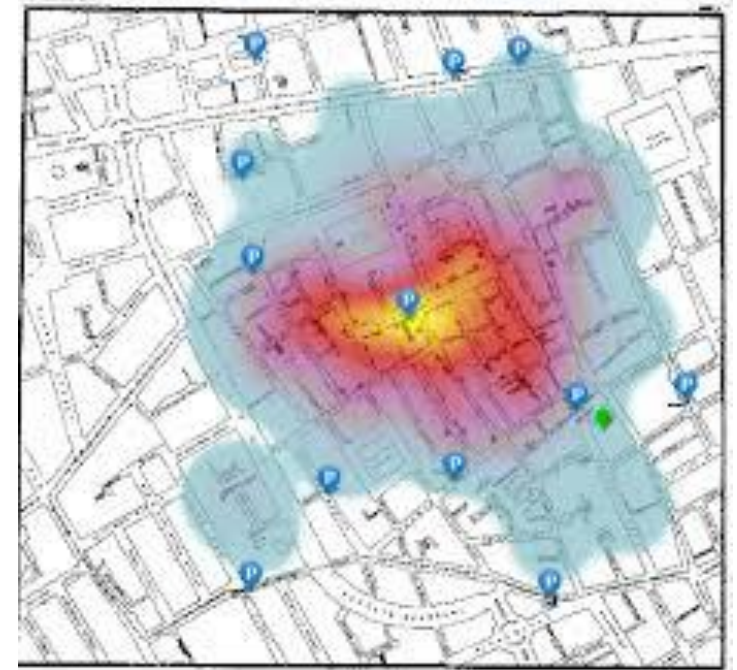
	POSITION	C	PF	PG	SF	SG
TEAM						
Atlanta Hawks		18.206896	23.500000	37.096500	9.835881	17.071120
Boston Celtics		26.500000	4.306281	22.800000	30.351780	17.142857
Brooklyn Nets		9.391069	44.119845	38.917057	20.100000	19.500000
Charlotte Bobcats		3.722040	1.563518	8.623920	30.075000	21.486316
Chicago Bulls		3.200000	7.775400	9.030000	27.300000	37.096500

Quick review of maps

Visualizing data on a map can be a powerful way to see spatial trends

We can create maps in Python using geopandas DataFrames

- Like regular DataFrames with an additional geometry column that has Shaply objects



John Snow's ghost map (1854)

	key_comb_drvr	geometry
0	M11551	POINT (117.525391 34.008926)
1	M17307	POINT (86.51248 30.474344)
2	M19584	POINT (89.537415 37.157627)

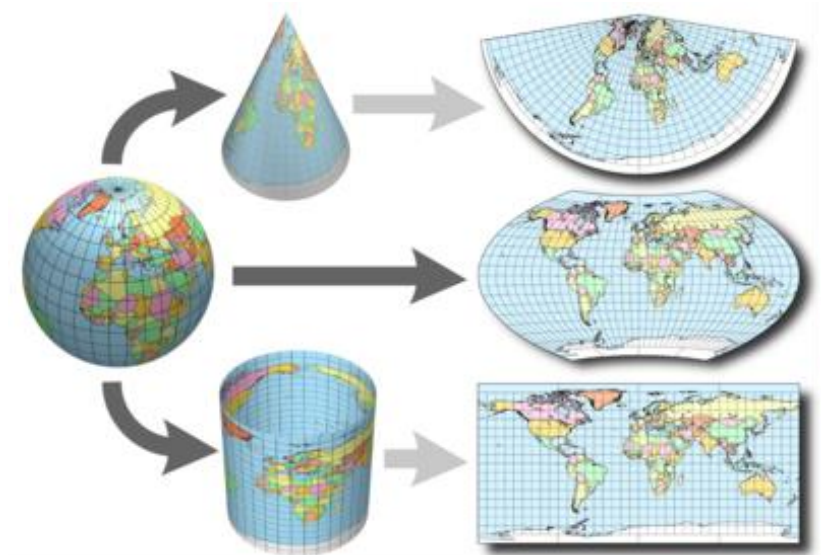
Review: CRSs and map projections

A coordinate reference system (CRS) is a framework used to precisely measure locations on the surface of the Earth as coordinates

- Needed for aligning different layers on maps

There are many map projections to display Earth's 3D structure on a 2D map surface.

- **Mercator projection** keeps angles intact
- **Eckert IV projection** keeps the size of land areas intact



WHAT YOUR FAVORITE MAP PROJECTION SAYS ABOUT YOU

MERCATOR



YOU'RE NOT REALLY INTO MAPS.

VAN DER GRINTEN



YOU'RE NOT A COMPLICATED PERSON. YOU LOVE THE MERCATOR PROJECTION; YOU JUST WISH IT WEREN'T SQUARE. THE EARTH'S NOT A SQUARE, IT'S A CIRCLE. YOU LIKE CIRCLES. TODAY IS GONNA BE A GOOD DAY!

HOBBO-DYER



YOU WANT TO AVOID CULTURAL IMPERIALISM, BUT YOU'VE HEARD BAD THINGS ABOUT GAIL-PETERS. YOU'RE CONFLICT-AVERSE AND BUY ORGANIC. YOU USE A RECENTLY-INVENTED SET OF GENDER-NEUTRAL PRONOUNS AND THINK THAT WHAT THE WORLD NEEDS IS A REVOLUTION IN CONSCIOUSNESS.

PLATE CARREE (EQUIRECTANGULAR)



YOU THINK THIS ONE IS FINE. YOU LIKE HOW X AND Y MAP TO LATITUDE AND LONGITUDE. THE OTHER PROJECTIONS OVERCOMPLICATE THINGS. YOU WANT ME TO STOP ASKING ABOUT MAPS SO YOU CAN ENJOY DINNER.

ROBINSON



YOU HAVE A COMFORTABLE PAIR OF RUNNING SHOES THAT YOU WEAR EVERYWHERE. YOU LIKE COFFEE AND ENJOY THE BEATLES. YOU THINK THE ROBINSON IS THE BEST-LOOKING PROJECTION, HANDS DOWN.

DYMAXION



YOU LIKE ISAAC ASIMOV, XML, AND SHOES WITH TOES. YOU THINK THE SEAWAY GOT A BAD RAP. YOU OWN 3D GOGGLES, WHICH YOU USE TO VIEW ROTATING MODELS OF BETTER 3D GOGGLES. YOU TYPE IN DVDRK.

A GLOBE!



YES, YOU'RE VERY CLEVER.

WATERMAN BUTTERFLY



REALLY? YOU KNOW THE WATERMAN? HAVE YOU SEEN THE 1909 CAHILL MAP ITS BASED— ... YOU HAVE A FRAMED REPRODUCTION AT HOME?! WHOA ... LISTEN. FORGET THESE QUESTIONS. ARE YOU DOING ANYTHING TONIGHT?

WINKEL-TRIEPEL



NATIONAL GEOGRAPHIC ADOPTED THE WINKEL-TRIEPEL IN 1998, BUT YOU'VE BEEN A WAT FAN SINCE LONG BEFORE 'NAT'GEO' SHOWED UP. YOU'RE WORRIED IT'S GETTING PLAYED OUT, AND ARE THINKING OF SWITCHING TO THE KAVRANSKY. YOU ONCE LEFT A PARTY IN DISGUST WHEN A GUEST SHOWED UP WEARING SHOES WITH TOES. YOUR FAVORITE MUSICAL GENRE IS "POST-".

GOODE HOMOLOGINE



THEY SAY MAPPING THE EARTH ON A 2D SURFACE IS LIKE FLATTENING AN ORANGE PEEL, WHICH SEEMS EASY ENOUGH TO YOU. YOU LIKE EASY SOLUTIONS. YOU THINK WE WOULDN'T HAVE SO MANY PROBLEMS IF WE'D JUST ELECT *ADORABLE* PEOPLE TO CONGRESS INSTEAD OF POLITICIANS. YOU THINK AIRLINES SHOULD JUST BUY FODD FROM THE RESTAURANTS NEAR THE GATES AND SERVE *JAM* ON BOARD. YOU CHANGE YOUR CAR'S OIL, BUT SECRETLY WONDER IF YOU REALLY *NEED* TO.

PEIRCE QUINCUNCIAL



YOU THINK THAT WHEN WE LOOK AT A MAP, WHAT WE REALLY SEE IS OURSELVES. PETER YOU FIRST SAW *INCEPTION*, YOU SAT SILENT IN THE THEATER FOR SIX HOURS. IT BREAKS YOU OUT TO REALIZE THAT EVERYONE AROUND YOU HAS A SKELETON INSIDE THEM. YOU *HAVE* REALLY LOOKED AT YOUR HANDS.

GAIL-PETERS



I HATE YOU.

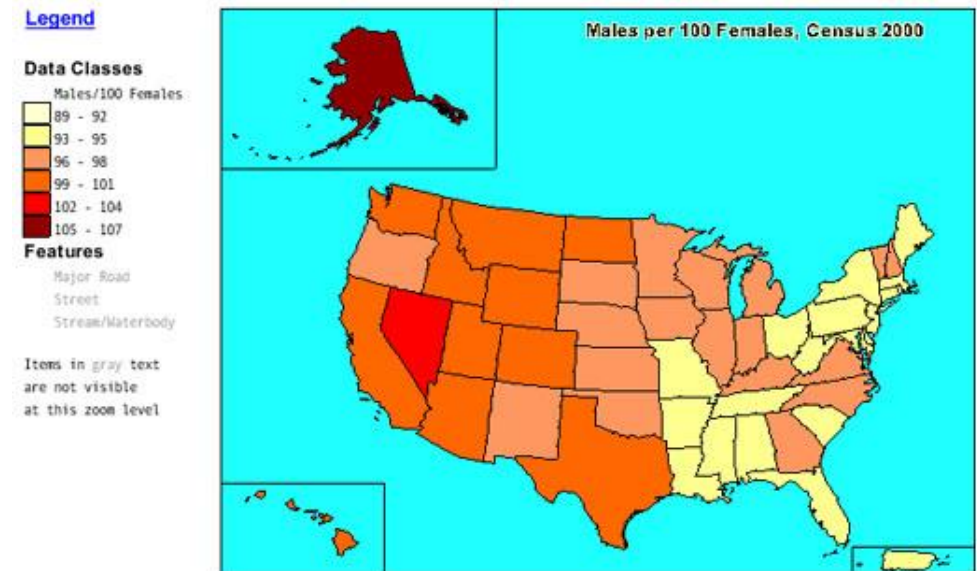
Choropleth maps

Choropleth maps: shades/colors in predefined areas based on properties of a variable

We can create choropleth maps using geopandas by joining region information on to a geopandas DataFrame that has a map

We can then use the `gpd.plot(column = )` method to visualize the map

Choropleth map



Let's explore this in Jupyter!

Loops

For loops

For loops repeat a process many times, iterating over a sequence of items

- Often we are iterating over an array of sequential numbers

```
animals = ["cat", "dog", "bat"]
```

```
for creature in animals:
```

```
    print(creature)
```

```
for i in np.arange(4):
```

```
    print(i**2)
```


Ranges

A range gives us a sequence of consecutive numbers

An sequence of increasing integers from 0 up to *end* - 1

- `range(end)`

An sequence of increasing integers from *start* up to *end* - 1

- `range(start, end)`

A sequence with step between consecutive values

- `range(start, end, step)`

The range always includes start but excludes end

Let's explore this in Jupyter!

Enumerate and zip

We can use the `enumerate()` function to both items in a list, and sequential integers:

```
animals = ["cat", "dog", "bat"]  
for i, creature in enumerate(animals):  
    print(i, creature)
```

We can use the `zip()` function to get items for two lists:

```
animal_order = ["feline", "canine", "chiropteran"]  
for curr_order, curr_animal in zip(animal_order, animals):  
    print(curr_order, curr_animal)
```

Let's explore this in Jupyter!

Conditional statements

Review: comparisons

We can use mathematical operators to compare numbers and strings

- Results return Boolean values **True** and **False**

Comparison	Operator	True example	False Example
Less than	<	2 < 3	2 < 2
Greater than	>	3 > 2	3 > 3
Less than or equal	<=	2 <= 2	3 <= 2
Greater or equal	>=	3 >= 3	2 >= 3
Equal	==	3 == 3	3 == 2
Not equal	!=	3 != 2	2 != 2

We can also make comparisons across elements in an array

Conditional statements

Conditional statements control the sequence of computations that are performed in a program

We use the keyword `if` to begin a conditional statement to only execute lines of code if a particular condition is met.

We can use `elif` to test additional conditions

We can use an `else` statement to run code if none of the `if` or `elif` conditions have been met.

```
num = 5
if num == 1:
    print("Monday")
elif num == 2:
    print("Tuesday")
elif num == 3:
    print("Wednesday")
elif num == 4:
    print("Thursday")
elif num == 5:
    print("Friday")
elif num == 6:
    print("Saturday")
elif num == 7:
    print("Sunday")
else:
    print("Invalid input")
```

Let's explore this in Jupyter!